OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) III - Semester (MAIN) Examinations Dec-2024/Jan 2025

OBJECT ORIENTED PROGRAMMING USING JAVA

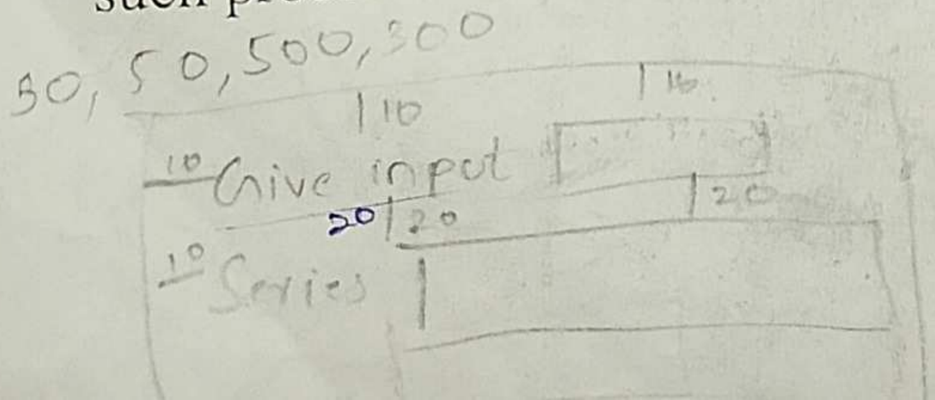
Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** is compulsory and answer any **four questions** from the remaining questions (2- 7), each Question carries 12 Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any, may suitably be assumed.***

		Marks	BT	CO
1. a)	How to implement inheritance in a child class. Write down the complete code with an example.	2	3	1
b)	How many ways can a thread be created. Write down the examples using code snippets.	2	1	2
c)	What is the important difference between the List implementations, ArrayList and Vector.	2	2	3
d)	What do you understand about the MVC pattern applicable to java swing GUI models	2	3	4
e)	Address is a POJO class representing the mailing address containing houseNo, streetName and city as fields. Employee class is also a POJO class which contains fields, name, id, designation and address, where address is the reference object of Address. Write down the Employee and Address classes.	2	4	5
f)	What is the ternary operator. Write down a code snippet with an example.	2	1	1
2. a)	What are the four principles of Object Oriented Programming. Explain each of them with an example.	6	2	1
b)	Write a Java Program to print factorial of **n** using Method Recursion. The value of number, **n** is provided within the program itself.	6	1	1
3. a)	Describe the Checked and Unchecked Exceptions by providing an example for each of them. Mention any 2 differences.	6	1	2
b)	Using the **PrintWriter** Class, develop a program to save the contents provided within the program itself.	6	2	2

(P.T.O.)

4. a)	Using **StringTokenizer** class, display all the tokens by splitting them into protocol, host-name, port , path and searchword of the given address. For example: String, address = "http://www.gallant.com:8080/Find?noble"; Format is: protocol: // hostname [: port] / path [? searchwords]	6	5	3
b)	Write down the interface hierarchy of the **Collection** interface using a neat sketch. Write down all of the concrete class implementations available in the hierarchy with their salient characteristics.	6	3	3
5. a)	Why are the Layout managers important in GUI development. Describe all of the Layout managers used in swing development.	6	3	4
b)	Using java swing GUI, develop a program to form a pattern of series of 5 terms where each term is separated by a comma(',') from the provided seed and the step/increment values. The 1st term is seed value and any of the next term is the sum of step value and the previous term. The GUI will have a total of 4 components: 2 Text Fields, a Button and a Label which should appear in the same order. The first TextField is meant for entering the seed and the second Text field is for step value. When the button is clicked, the series of the pattern should be displayed in the label. [For Example, seed is 8 and the step is 4, the pattern would look like as: 8, 12, 16, 20, 24]	6	4	4
6. a)	Develop a program to serialize the Student Object state. Address is a POJO Class with the fields: HNo, streetName, city, pin. A Student is also a POJO Class with the fields: name, RNo, specialization, address where address is the reference object of Address. Assume the applicable data types.	6	5	5
b)	What is an Object Graph that is applicable to serialization. How is it achieved that the given object variables NOT get serialized from the plain object.	6	3	5
7. a)	Write down the interface hierarchy of the **Collection** interface using a neat sketch. Write down all of the concrete class implementations available in the hierarchy with their salient characteristics.	6	1	3
b)	What do you understand about Concurrency problems. How can such problems be addressed.	6	1	1



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